Thus the selected double glazing saves the following cost over :

yuan.

4.

. First let one electron tunnel from plate to . Let be the charge on plate before tunnelling, and the plate voltage (problem Fig. ). Then

The stored capacitor energy is

After the electron tunnels to , the charge on is

Here is electron charge magnitude. The new voltage and stored energy are

Coulomb blockade, forbidding tunnelling, requires

Equations give

Now let an electron tunnel from to . Again is the charge on before tunnelling, the plate voltage. When it tunnels from to , plate has charge . Similarly, forbidding tunnelling from to requires the following condition:

From , voltage between blocks single-electron tunnelling. The blockade condition is

. By the stated condition and , the tunnelling threshold satisfies

Substituting the data into gives